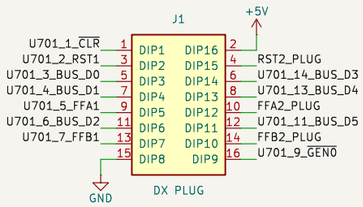
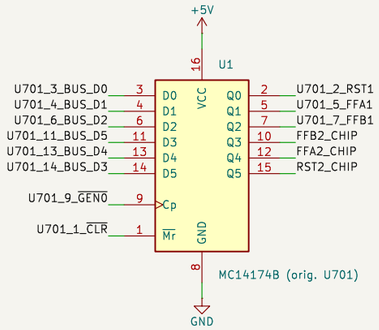
DX TimeShift rev A — snare delay interposer for the Oberheim DX

Fits Oberheim DX processor board 1721A (U701). NOT for DXa (board 1906, U715: snare lines are on U715 pins 7/5/2).
UNPLUG NANO BEFORE USB: flash the Nano OUT of the board, or only with the DX switched on (the Nano 5V pin is the DX logic rail).
Implements PROJECT_BRIEF.md 3.5 (LOCKED). Signal facts: docs/research/dx-findings.md sections b, c, d, h.

A. DX socket interface (J1 = plug to DX, U1 = original U701)

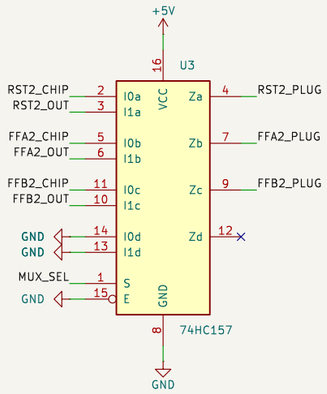




J1 pin number = box-header pin = ribbon conductor.
J1 pin NAME (DIPn) = DIP-16 plug pin = DX U701 socket pin.
IDC mapping (header->DIP): 1->1 2->16 3->2 4->15 5->3 6->14
7->4 8->13 9->5 10->12 11->6 12->11 13->7 14->10 15->8 16->9.
TO BE CONFIRMED with a continuity meter before first power-up.
DIP 10/12/15 (snare) come from the mux U3, never from U1.

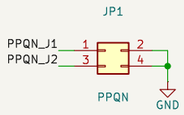
U1 = DIP-16 SOCKET. Fit the ORIGINAL U701
(MC14174B / 74C174) taken out of the DX.
Notch/pin 1 as marked on the PCB.
Pins 10/12/15 (FFB2/FFA2/RST2) are intercepted;
all other pins go straight to J1 (same DIP pin).

B. Bypass / delay multiplexer

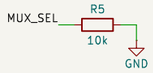


74HC157: I0x = input A (bypass), I1x = input B (delayed),
Zx = output Y. S LOW = A = bypass (R5 pulls S low when the Nano
is off / in reset / unprogrammed). S HIGH = B = delayed replay.
Channel 1 = RST2, 2 = FFA2, 3 = FFB2, 4 unused (inputs to GND).

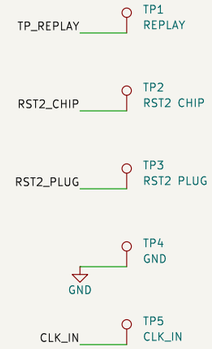
C. PPQN jumper, mux pull-down, test points



PPQN (Nano D10/D11, internal pull-ups):
JP1 both open = 96 (default, plain DX)
JP1 1-2 closed (D10) = 48
JP1 3-4 closed (D11) = 24 (DXa/Stretch/MIDI slaved)



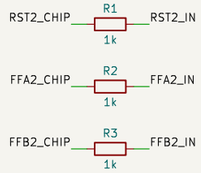
R5: MUX_SEL pull-down
= bypass by default

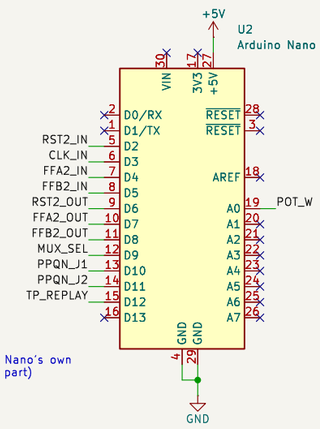


Test points (scope):
TP1 replay pulse, TP2 RST2 chip side,
TP3 RST2 plug side, TP4 GND, TP5 clock

D. Arduino Nano (U2) on 2 x 1x15 female headers

Series protection 1k
(U1 outputs -> Nano inputs)



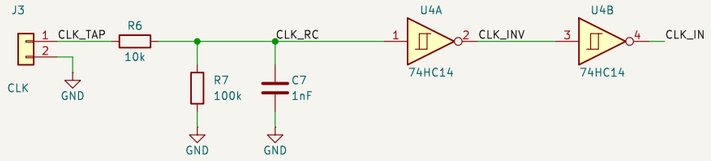


D13 = LED_ACT: the Nano's own
'L' LED (no external part)

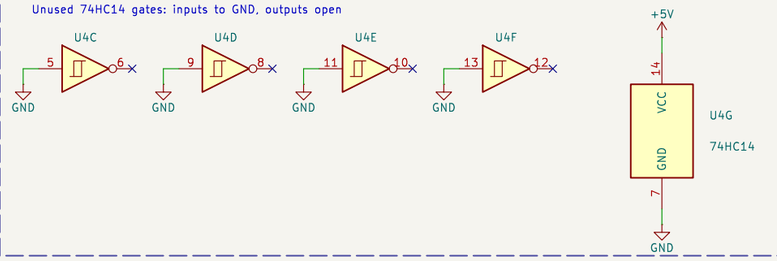
U2 +5V is the DX logic rail (via J1 DIP16).
UNPLUG NANO BEFORE USB.

E. DX clock input (96 PPQN tick, rising edge)

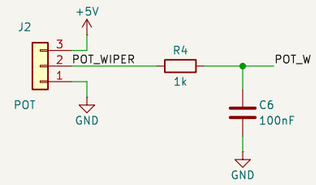
J3 pin 1 -> DX CLOCK OUT node: R791 lead on the Q709-collector side
(0/5 V, idles low, rising edge = tick). Clip-on grabber or tack solder.
R6 10k series + R7 100k pull-down + C7 1nF -> two Schmitt inverters
in series = non-inverting: rising edge at R791 = rising edge at D3.



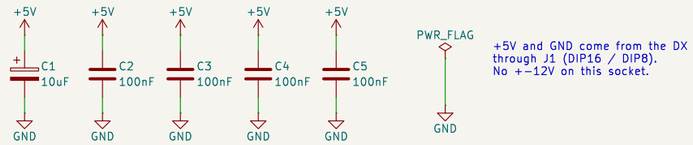
Unused 74HC14 gates: inputs to GND, outputs open



F. SNARE DELAY pot, power, decoupling, mounting



J2 -> rear-panel 10k lin pot "SNARE DELAY 0 <-> 1 BEAT":
1 = CCW lug (GND), 2 = wiper, 3 = CW lug (+5V).
R4 + C6 = RC filter into A0.



+5V and GND come from the DX
through J1 (DIP16 / DIP8).
No +-12V on this socket.

C1 bulk at J1; C2 at U1, C3 at U3, C4 at U4 pin 14, C5 at Nano pins 27/29



Generated: hardware/kicad/tools/gen_schematic.py
UNPLUG NANO BEFORE USB (flash it out of the board)
Fits Oberheim DX board 1721A (U701). NOT for DXa (U715).
DX TimeShift project

Sheet: /
File: timeshift.kicad_sch

Title: DX TimeShift rev A — snare delay interposer

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1